

Roll No.

TMA-202

B. TECH. (BIOTECH.) (SECOND SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

ENGINEERING MATHEMATICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) In the city of Dehradun there are 50 colleges and 100 schools. Each school and college has 1 peon, 5 clerks, 1 cashier. Each college, in addition has 1 accountant and 1 head clerk. The monthly salary of each of them is as follows :

Peon ₹ 150, clerk ₹ 250, cashier ₹ 300, accountant ₹ 350, head clerk ₹ 400.

Determine :

(CO1)

- (i) The total number of posts of each kind in school and colleges taken together.
- (ii) The total monthly salary bill of each school and college separately.
- (iii) The total monthly salary bill of all schools and colleges taken together.

P. T. O.

OR

- (b) A firm produces 3 product X, Y and Z requiring mix up of 3 materials A, B and C. Per unit requirement of each product for each material is given below : (CO1)

	X	Y	Z
A	2	5	3
B	4	2	2
C	3	2	3

Determine :

- (i) Total requirement of each material if the firm produces 100, 200 and 300 units of the three products respectively.
 - (ii) Per unit cost of production if per unit cost of materials is ₹ 10, ₹ 5 and ₹ 8 respectively.
 - (iii) Total cost of production
2. (a) Three firms A, B and C supplied 40, 35 and 25 trucks loads of stones and 10, 5 and 8 truck loads of sand respectively to a contractor. If the costs of stones and sand are ₹ 1,200 and ₹ 500 per truck load respectively, find the total amount paid by the contractor to each of these firms. (CO1)

OR

- (b) Find the eigen values and eigen vectors of the following matrix :

(CO1)

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

(3)

3. (a) Solve the system of equations by matrix inversion method : (CO1)

$$x + y + z = 6$$

$$x + 2y + 3z = 14$$

$$-x + y - z = -2$$

OR

- (b) Solve the system of equations by elementary row operation : (CO1)

$$x + 2y - 2z = -1$$

$$3x + y + 4z = 17$$

$$5x - 3y + z = 2$$

4. (a) Investigate the maximum and minimum values of the function : (CO2)

$$3x^3 - 36x^2 + 135x - 13$$

OR

- (b) If:

$$y = x^3 \log \frac{1}{x}$$

prove that :

(CO2)

$$x \frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + 3x^2 = 0.$$

5. (a) Find $\frac{dy}{dx}$ when :

(CO2)

$$y = \frac{\sqrt{(5x+8)} \sqrt[3]{(1-2x)}}{\sqrt[4]{(x^2+10x)}}.$$

OR

- (b) Evaluate :

(CO2)

$$\int x^2 \log x \, dx \text{ and } \int x^7 e^{5x} \, dx$$

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TCH-202

**B. TECH. (BIOTECH.) (SECOND SEMESTER)
MID SEMESTER EXAMINATION, March, 2024**

ADVANCED ORGANIC CHEMISTRY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Discuss the Sulphonation reaction of benzene with mechanism. (CO2)

OR

- (b) Elaborate the Friedel-Craft's alkylation reaction with mechanism. (CO2)

2. (a) Explain the terms electrophiles and nucleophiles with suitable example. (CO2)

OR

- (b) Evaluate the S_N^1 reaction with mechanism and explain which alkyl halide give S_N^1 reaction faster. (CO2)

3. (a) Define the following terms : (CO2)

(i) Resonance

(ii) Electromeric and Inductive effect

OR

- (b) Explain the term free radical in terms of hybridization and stability.

(CO5)

P. T. O.

(2)

4. (a) Discuss the preparation, physical and chemical properties of glucose.

(CO5)

OR

- (b) Evaluate the importance of the Killiani-Fischer synthesis in carbohydrate chemistry.

(CO5)

5. (a) Discuss the Ruff degradation reaction.

(CO5)

OR

- (b) Discuss in brief introduction about the carbohydrate and give its classification with suitable example.

(CO5)